```
import pandas as pd
import numpy as np

df = pd.read_csv("/content/data.csv")
print("Shape:", df.shape)
df.head()
```

```
Shape: (150000, 12)
```

	Unnamed: 0	SeriousDlqin2yrs	RevolvingUtilizationOfUnsecuredLines	age	NumberOfTime30-59DaysPastDueNotWorse	DebtRatio	MonthlyIncome	N
0	1	1	0.766127	45	2	0.802982	9120.0	
1	2	0	0.957151	40	0	0.121876	2600.0	
2	3	0	0.658180	38	1	0.085113	3042.0	
3	4	0	0.233810	30	0	0.036050	3300.0	
4	5	0	0.907239	49	1	0.024926	63588.0	

Credit Risk Prediction (Give Me Some Credit) with XAI (SHAP)

Dataset: Give Me Some Credit (Kaggle)
Goal: Predict `SeriousDlqin2yrs` (1 = default) and explain predictions using SHAP.

Steps:

1. Load & explore data
2. Handle missing values, scaling
3. Deal with class imbalance (SMOTE)
4. Train XGBoost classifier
5. Evaluate with ROC-AUC, confusion matrix
6. Explain model using SHAP (global + local)

```
import numpy as np
import pandas as pd

import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.pipeline import Pipeline
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import (
    roc_auc_score,
    classification_report,
    confusion_matrix,
    roc_curve
)

from imblearn.over_sampling import SMOTE

import xgboost as xgb
import shap
import joblib

# For nicer plots
sns.set(style="whitegrid")
plt.rcParams["figure.figsize"] = (8, 5)
```

```
DATA_PATH = "/content/cs-training.csv"  # <-- change if your path is different

df = pd.read_csv(DATA_PATH)
print("Shape:", df.shape)
df.head()
```

```
Shape: (150000, 12)
```

	Unnamed: 0	SeriousDlqin2yrs	RevolvingUtilizationOfUnsecuredLines	age	NumberOfTime30-59DaysPastDueNotWorse	DebtRatio	MonthlyIncome	N
0	1	1	0.766127	45	2	0.802982	9120.0	
1	2	0	0.957151	40	0	0.121876	2600.0	
2	3	0	0.658180	38	1	0.085113	3042.0	
3	4	0	0.233810	30	0	0.036050	3300.0	
4	5	0	0.907239	49	1	0.024926	63588.0	

```
df.columns
```

```
Index(['Unnamed: 0', 'SeriousDlqin2yrs',
       'RevolvingUtilizationOfUnsecuredLines', 'age',
       'NumberOfTime30-59DaysPastDueNotWorse', 'DebtRatio', 'MonthlyIncome',
       'NumberOfOpenCreditLinesAndLoans', 'NumberOfTimes90DaysLate',
       'NumberRealEstateLoansOrLines', 'NumberOfTime60-89DaysPastDueNotWorse',
       'NumberOfDependents'],
      dtype='object')
```

```
# Drop useless index-like column
if "Unnamed: 0" in df.columns:
    df = df.drop(columns=["Unnamed: 0"])

# Separate target
target_col = "SeriousDlqin2yrs"
X = df.drop(columns=[target_col])
y = df[target_col]

print("New shape:", X.shape)
print("Target distribution:")
print(y.value_counts())
print("\nTarget distribution (normalized):")
print(y.value_counts(normalize=True))
```

```
New shape: (150000, 10)
Target distribution:
SeriousDlqin2yrs
0    139974
1     10026
Name: count, dtype: int64

Target distribution (normalized):
SeriousDlqin2yrs
0    0.93316
1    0.06684
Name: proportion, dtype: float64
```

```
df.isnull().sum()
```

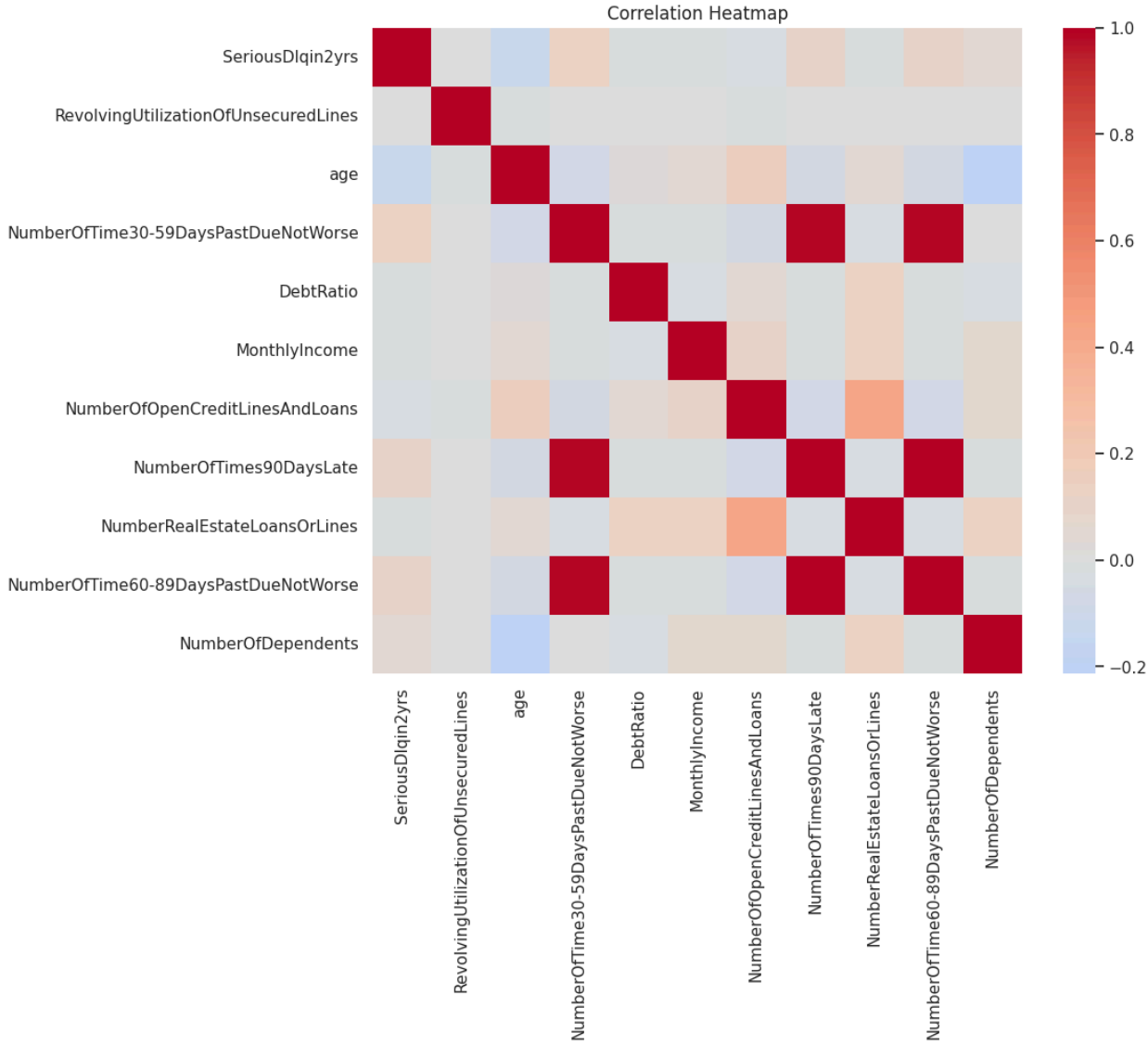
	0
SeriousDlqin2yrs	0
RevolvingUtilizationOfUnsecuredLines	0
age	0
NumberOfTime30-59DaysPastDueNotWorse	0
DebtRatio	0
MonthlyIncome	29731
NumberOfOpenCreditLinesAndLoans	0
NumberOfTimes90DaysLate	0
NumberRealEstateLoansOrLines	0
NumberOfTime60-89DaysPastDueNotWorse	0
NumberOfDependents	3924

dtype: int64

```
df.describe().T
```

	count	mean	std	min	25%	50%	75%	max
SeriousDlqin2yrs	150000.0	0.066840	0.249746	0.0	0.000000	0.000000	0.000000	1.0
RevolvingUtilizationOfUnsecuredLines	150000.0	6.048438	249.755371	0.0	0.029867	0.154181	0.559046	50708.0
age	150000.0	52.295207	14.771866	0.0	41.000000	52.000000	63.000000	109.0
NumberOfTime30-59DaysPastDueNotWorse	150000.0	0.421033	4.192781	0.0	0.000000	0.000000	0.000000	98.0
DebtRatio	150000.0	353.005076	2037.818523	0.0	0.175074	0.366508	0.868254	329664.0
MonthlyIncome	120269.0	6670.221237	14384.674215	0.0	3400.000000	5400.000000	8249.000000	3008750.0
NumberOfOpenCreditLinesAndLoans	150000.0	8.452760	5.145951	0.0	5.000000	8.000000	11.000000	58.0
NumberOfTimes90DaysLate	150000.0	0.265973	4.169304	0.0	0.000000	0.000000	0.000000	98.0
NumberRealEstateLoansOrLines	150000.0	1.018240	1.129771	0.0	0.000000	1.000000	2.000000	54.0
NumberOfTime60-89DaysPastDueNotWorse	150000.0	0.240387	4.155179	0.0	0.000000	0.000000	0.000000	98.0

```
# Correlation heatmap (for top correlations)
plt.figure(figsize=(10, 8))
corr = df.corr(numeric_only=True)
sns.heatmap(corr, annot=False, cmap="coolwarm", center=0)
plt.title("Correlation Heatmap")
plt.show()
```




```
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

X_train.shape, X_test.shape
```

```
((120000, 10), (30000, 10))
```

```
numeric_features = X.columns.tolist()
numeric_features
```

```
['RevolvingUtilizationOfUnsecuredLines',
 'age',
 'NumberOfTime30-59DaysPastDueNotWorse',
 'DebtRatio',
 'MonthlyIncome',
 'NumberOfOpenCreditLinesAndLoans',
 'NumberOfTimes90DaysLate',
 'NumberRealEstateLoansOrLines',
 'NumberOfTime60-89DaysPastDueNotWorse',
 'NumberOfDependents']
```

```
num_pipeline = Pipeline([
    ("imputer", SimpleImputer(strategy="median")),
    ("scaler", StandardScaler())
])

# Fit on train, transform both
X_train_prep = num_pipeline.fit_transform(X_train)
X_test_prep = num_pipeline.transform(X_test)

X_train_prep.shape, X_test_prep.shape
```

```
((120000, 10), (30000, 10))
```

```
print("Class counts before SMOTE:", np.bincount(y_train))

smote = SMOTE(random_state=42)
X_train_bal, y_train_bal = smote.fit_resample(X_train_prep, y_train)

print("Class counts after SMOTE:", np.bincount(y_train_bal))
```

```
Class counts before SMOTE: [111979   8021]
Class counts after SMOTE: [111979 111979]
```


```
print("Class counts before SMOTE:", np.bincount(y_train))

smote = SMOTE(random_state=42)
X_train_bal, y_train_bal = smote.fit_resample(X_train_prep, y_train)

print("Class counts after SMOTE:", np.bincount(y_train_bal))
```

```
Class counts before SMOTE: [111979   8021]
Class counts after SMOTE: [111979 111979]
```


```
model = xgb.XGBClassifier(
    n_estimators=300,
    learning_rate=0.05,
    max_depth=5,
    subsample=0.8,
    colsample_bytree=0.8,
    random_state=42,
    eval_metric="logloss",
    tree_method="hist"  # faster; remove if error on your GPU/CPU setup
)

model.fit(X_train_bal, y_train_bal)
```

```
XGBClassifier
XGBClassifier(base_score=None, booster=None, callbacks=None,
              colsample_bylevel=None, colsample_bynode=None,
              colsample_bytree=0.8, device=None, early_stopping_rounds=None,
              enable_categorical=False, eval_metric='logloss',
              feature_types=None, feature_weights=None, gamma=None,
              grow_policy=None, importance_type=None,
              interaction_constraints=None, learning_rate=0.05, max_bin=None,
              max_cat_threshold=None, max_cat_to_onehot=None,
              max_delta_step=None, max_depth=5, max_leaves=None,
              min_child_weight=None, missing=nan, monotone_constraints=None,
              multi_strategy=None, n_estimators=300, n_jobs=None,
```

```
# Predictions
y_pred_proba = model.predict_proba(X_test_prep)[:, 1]
y_pred = model.predict(X_test_prep)

# ROC-AUC
auc = roc_auc_score(y_test, y_pred_proba)
print("ROC-AUC:", auc)
```

```
ROC-AUC: 0.8464413533054308
```

```
print("Classification report:")
print(classification_report(y_test, y_pred))
```

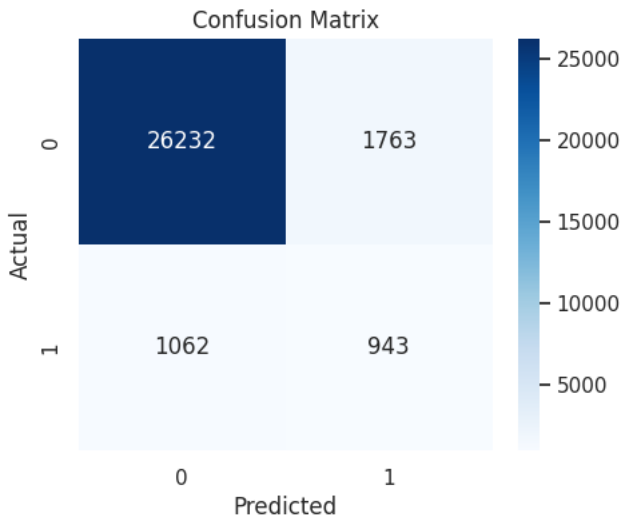
```
Classification report:
              precision    recall  f1-score   support
```

```
           0       0.96      0.94      0.95     27995
           1       0.35      0.47      0.40      2005

    accuracy                           0.91     30000
   macro avg       0.65      0.70      0.67     30000
weighted avg       0.92      0.91      0.91     30000
```

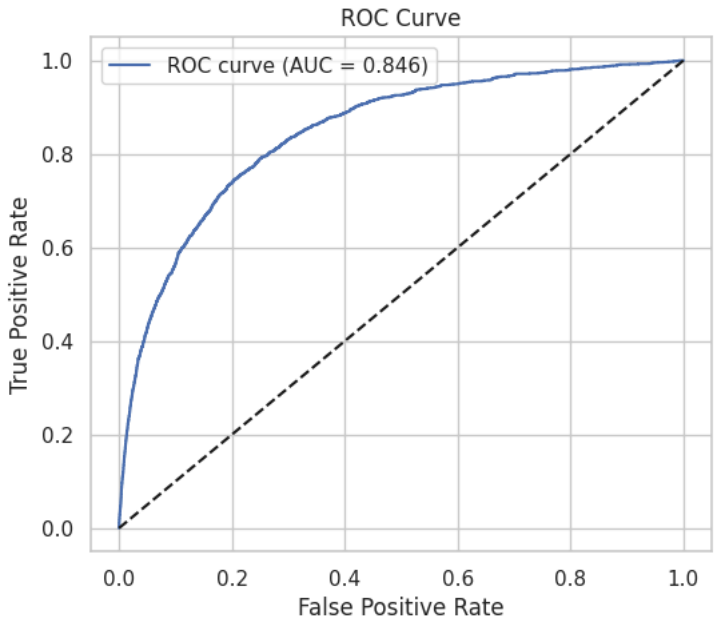
```
cm = confusion_matrix(y_test, y_pred)
cm
```

```
array([[26232,  1763],
       [ 1062,   943]])
```

```
# Plot confusion matrix
plt.figure(figsize=(5, 4))
sns.heatmap(cm, annot=True, fmt="d", cmap="Blues")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()
```




```
# ROC Curve
fpr, tpr, thresholds = roc_curve(y_test, y_pred_proba)
plt.figure(figsize=(6, 5))
plt.plot(fpr, tpr, label=f"ROC curve (AUC = {auc:.3f})")
plt.plot([0, 1], [0, 1], "k--")
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve")
plt.legend()
plt.show()
```

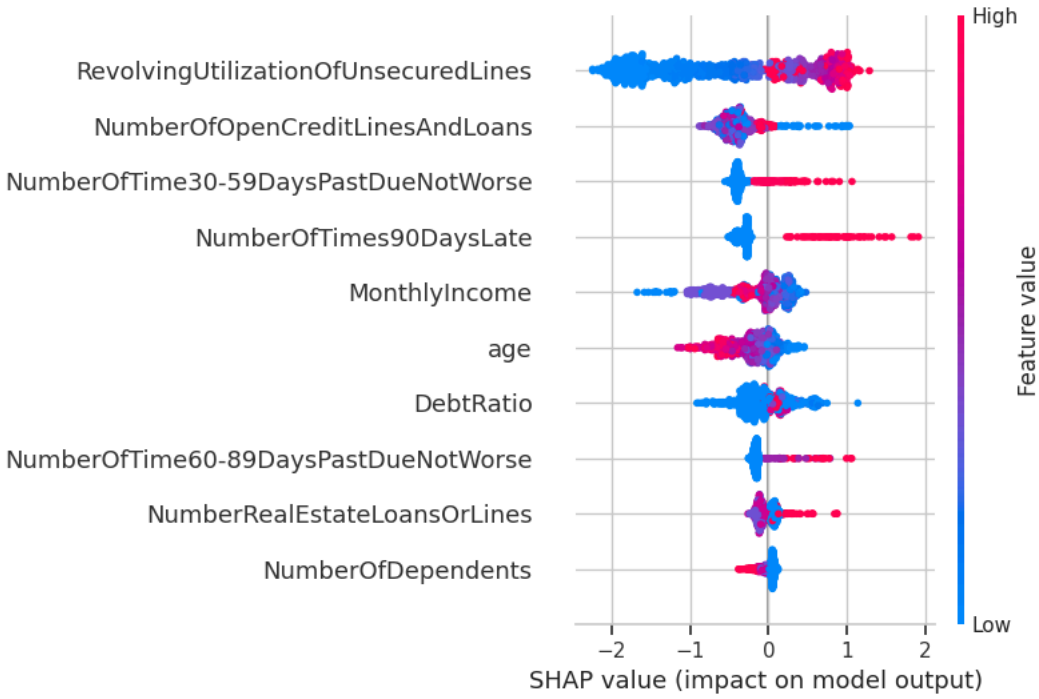



```
joblib.dump(model, "xgb_gmsc_model.joblib")
joblib.dump(num_pipeline, "gmsc_preprocessor.joblib")

print("Saved: xgb_gmsc_model.joblib and gmsc_preprocessor.joblib")
```

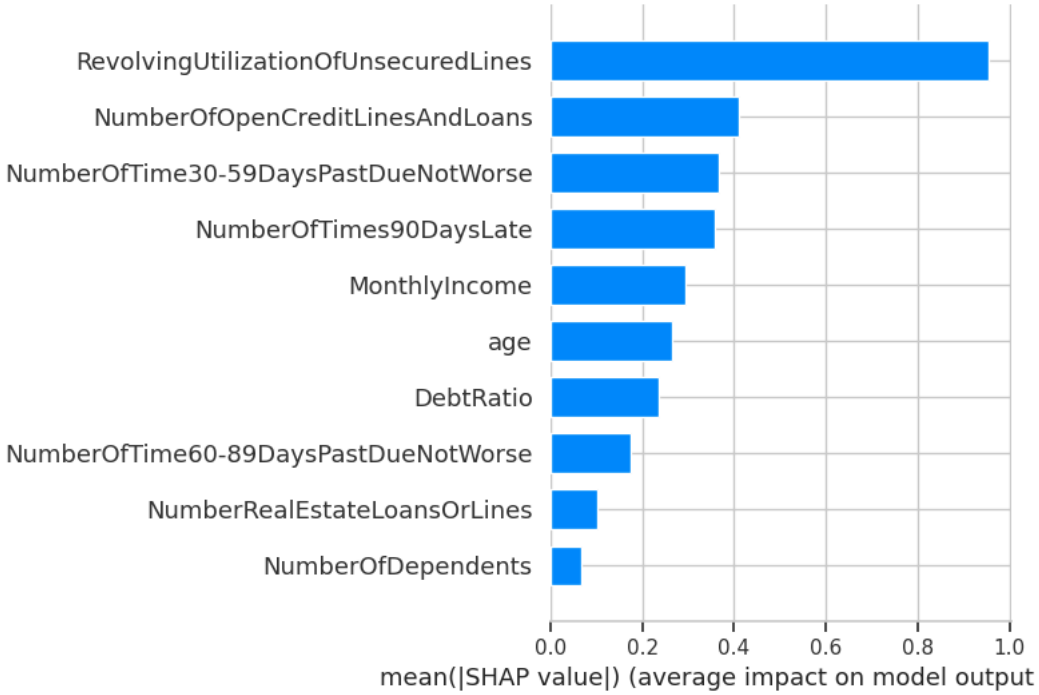
```
Saved: xgb_gmsc_model.joblib and gmsc_preprocessor.joblib
```

```
# Initialize JS if you're in Jupyter Notebook
shap.initjs()

# Feature names
feature_names = X.columns.tolist()

# Build explainer for the trained model
explainer = shap.TreeExplainer(model)

# Use a subset for SHAP to keep it fast
X_test_df = pd.DataFrame(X_test_prep, columns=feature_names)
X_shap = X_test_df.sample(1000, random_state=42)

# Compute SHAP values
shap_values = explainer.shap_values(X_shap)
```

```
shap.summary_plot(shap_values, X_shap)
```




```
shap.summary_plot(shap_values, X_shap, plot_type="bar")
```

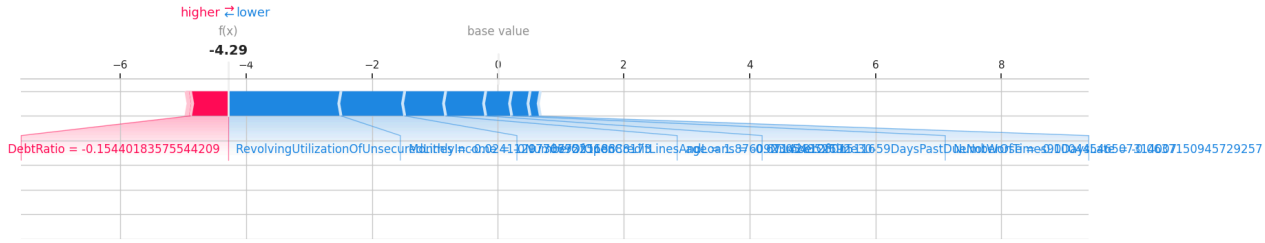



```
# Pick any instance from the test set
idx = 5  # change index as you like

instance = X_test_df.iloc[[idx]]          # preprocessed features
raw_instance = X_test.iloc[[idx]]         # original values for human reading

display(raw_instance)  # show original feature values
```

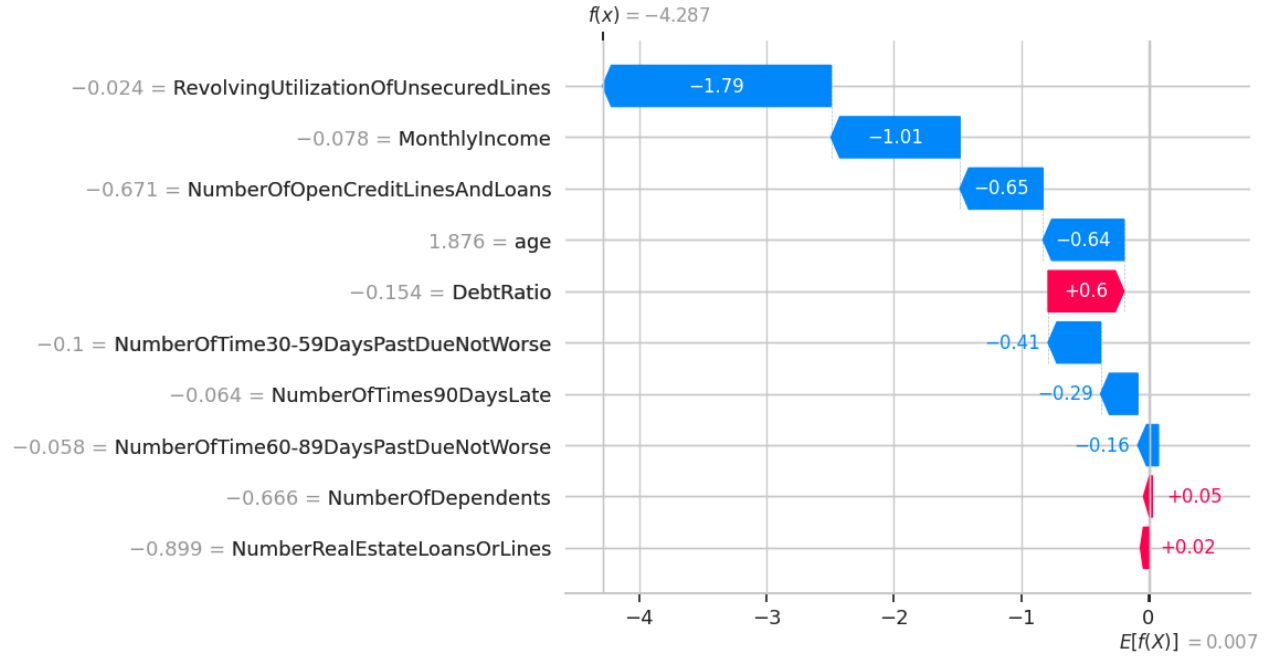
	RevolvingUtilizationOfUnsecuredLines	age	NumberOfTime30-59DaysPastDueNotWorse	DebtRatio	MonthlyIncome	NumberOfOpenCreditLinesAr
86739	0.019643	80	0	29.0	NaN	

```
# SHAP values for that one instance
shap_value_single = explainer.shap_values(instance)
```

```
# Force plot with matplotlib (static)
shap.force_plot(
    explainer.expected_value,
    shap_value_single[0, :],
    instance,
    matplotlib=True
)
```




```
try:
    exp = shap.Explanation(
        values=shap_value_single[0],
        base_values=explainer.expected_value,
        data=instance.values[0],
        feature_names=feature_names
    )
    shap.plots.waterfall(exp)
except Exception as e:
    print("Waterfall plot not supported in this SHAP version:", e)
```




```
mean_abs_shap = pd.DataFrame({
    "feature": feature_names,
    "mean_abs_shap": np.abs(shap_values).mean(axis=0)
}).sort_values(by="mean_abs_shap", ascending=False)

mean_abs_shap.head(10)
```

	feature	mean_abs_shap
0	RevolvingUtilizationOfUnsecuredLines	0.955403
5	NumberOfOpenCreditLinesAndLoans	0.410969
2	NumberOfTime30-59DaysPastDueNotWorse	0.365205
6	NumberOfTimes90DaysLate	0.357269
4	MonthlyIncome	0.294632
1	age	0.265233
3	DebtRatio	0.235730
8	NumberOfTime60-89DaysPastDueNotWorse	0.174878
7	NumberRealEstateLoansOrLines	0.101635
9	NumberOfDependents	0.068487

```
fair_df = X_test.copy()
fair_df["y_true"] = y_test.values
fair_df["y_pred"] = y_pred
fair_df["y_prob"] = y_pred_proba

# Age groups
fair_df["age_group"] = pd.cut(
    fair_df["age"],
    bins=[0, 30, 45, 60, 120],
    labels=["Young", "Adult", "Middle-Aged", "Senior"]
)

# Income groups
fair_df["income_group"] = pd.qcut(
    fair_df["MonthlyIncome"],
    q=4,
    labels=["Low", "Mid-Low", "Mid-High", "High"]
)

# Debt Ratio groups
fair_df["debt_group"] = pd.qcut(
    fair_df["DebtRatio"],
    q=4,
    labels=["Low Debt", "Mid-Low Debt", "Mid-High Debt", "High Debt"]
)

fair_df.head()
```

	RevolvingUtilizationOfUnsecuredLines	age	NumberOfTime30-59DaysPastDueNotWorse	DebtRatio	MonthlyIncome	NumberOfOpenCreditLinesA
10682	0.019252	66	0	0.433926	5175.0	
40173	0.454221	78	0	0.873533	2300.0	
4719	0.077572	59	0	0.172882	10833.0	
81730	0.107864	59	0	0.453692	9900.0	
113639	0.101095	67	0	0.407831	5950.0	

```
# Default rate by age group
fair_df.groupby("age_group")["y_true"].mean()
```

```
/tmp/ipython-input-3682778265.py:2: FutureWarning: The default of observed=False is deprecated and will be changed to True i
  fair_df.groupby("age_group")["y_true"].mean()
```

age_group	y_true
Young	0.128531
Adult	0.094969
Middle-Aged	0.066598
Senior	0.027255

dtype: float64

```
# Default rate by income group
fair_df.groupby("income_group")["y_true"].mean()
```

```
/tmp/ipython-input-800061622.py:2: FutureWarning: The default of observed=False is deprecated and will be changed to True ir
  fair_df.groupby("income_group")["y_true"].mean()
```

income_group	y_true
Low	0.094037
Mid-Low	0.077347
Mid-High	0.060986
High	0.044103

dtype: float64

```
# Default rate by debt group
fair_df.groupby("debt_group")["y_true"].mean()
```

```
/tmp/ipython-input-1779578097.py:2: FutureWarning: The default of observed=False is deprecated and will be changed to True i
  fair_df.groupby("debt_group")["y_true"].mean()
```

debt_group	y_true
Low Debt	0.059867
Mid-Low Debt	0.054000
Mid-High Debt	0.083600
High Debt	0.069867

dtype: float64

```
def group_recall(df, group_col):
    results = {}
    for g in df[group_col].dropna().unique():
        sub = df[df[group_col] == g]
        tp = ((sub["y_true"] == 1) & (sub["y_pred"] == 1)).sum()
        fn = ((sub["y_true"] == 1) & (sub["y_pred"] == 0)).sum()
        recall = tp / (tp + fn) if (tp + fn) > 0 else 0
        results[g] = recall
    return pd.Series(results)

group_recall(fair_df, "age_group")
```

	0
Senior	0.256098
Middle-Aged	0.431373
Young	0.652015
Adult	0.510363

dtype: float64

```
group_recall(fair_df, "income_group")
```

	0
Mid-Low	0.526998
Low	0.552398
High	0.287879
Mid-High	0.394521

dtype: float64

```
group_recall(fair_df, "debt_group")
```

	0
Mid-High Debt	0.400319
Low Debt	0.585746
High Debt	0.486641
Mid-Low Debt	0.429630

dtype: float64

```
def false_positive_rate_by_group(df, group_col):
    results = {}
    for g in df[group_col].dropna().unique():
        sub = df[df[group_col] == g]
        fp = ((sub["y_true"] == 0) & (sub["y_pred"] == 1)).sum()
        tn = ((sub["y_true"] == 0) & (sub["y_pred"] == 0)).sum()
        fpr = fp / (fp + tn) if (fp + tn) > 0 else 0
        results[g] = fpr
    return pd.Series(results)

false_positive_rate_by_group(fair_df, "age_group")
```

	0
Senior	0.009567
Middle-Aged	0.053163
Young	0.222582
Adult	0.099905

dtype: float64

```
false_positive_rate_by_group(fair_df, "income_group")
```